Sentiment Analysis and Topic Modeling With Machine Learning on Airline Reviews

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Introduction:

This project applies sentiment analysis, topic modeling, and machine learning classifications to analyze 129,455 airline customer reviews. The goal is to understand customer satisfaction patterns and to predict whether a customer would recommend an airline based on text alone.

Airline reviews contain important information about the passenger's experience, emotion, comfort, and overall satisfaction. By using text mining methods, airlines can understand the concerns of customers and improve the quality based on text analysis.

This project will focus on analyzing how sentiment score differs between recommended and non-recommended airlines, underlying context and themes utilizing LDA topic modeling, and whether machine learning models can accurately predict customer recommendations solely on text alone.

Data:

Dataset Details:

- Source: Airline Reviews
- Total Number of Reviews: 129,455
- CSV File with 22 columns
- Prediction Target: Recommended (Yes/No)

Variables to Consider:

- Reviews: Contains a full-text review from a customer
- Recommended: Whether the customer recommends the airline
- CabinType: Economy, Business, First Class, Premium Economy

Methodology

During the text preprocessing and feature engineering process, the review text was prepared before analysis. Missing reviews were replaced with an empty string to preserve the dataset. A train/test split of 70/30 was performed for the machine learning classification.

For machine learning models, the text was converted using CounterVectorizer to transform reviews into word frequency counts and TF-IDF Transformer to assign importance to meaningful words and reduce the common words.

Sentiment Analysis:

- Used to measure each sentiment for each review
- Categorized scores as Negative Score, Neutral Score, Positive Score, and Compound
  - (Overall sentiment from -1 to 1)

Topic Modeling:

- Used to define underlying themes from processed corpus('Review')
- Trial & error over 3 LDA Models (K = 5, 7, 10)
  - Utilized coherence scores and probabilities

Machine Learning Classifications:

- Three types of models used to predict airline recommendations:
  - Naive Bayes
  - Logistic Regression
  - Random Forest

Rationale for Method Selection:

- Why Sentiment Analysis?
  - Airline reviews have strong emotional language
  - Useful for identifying satisfaction trends across airlines
- Why Topic Modeling?
  - Creates actionable underlying insights, typically hidden due sheer volume of documents

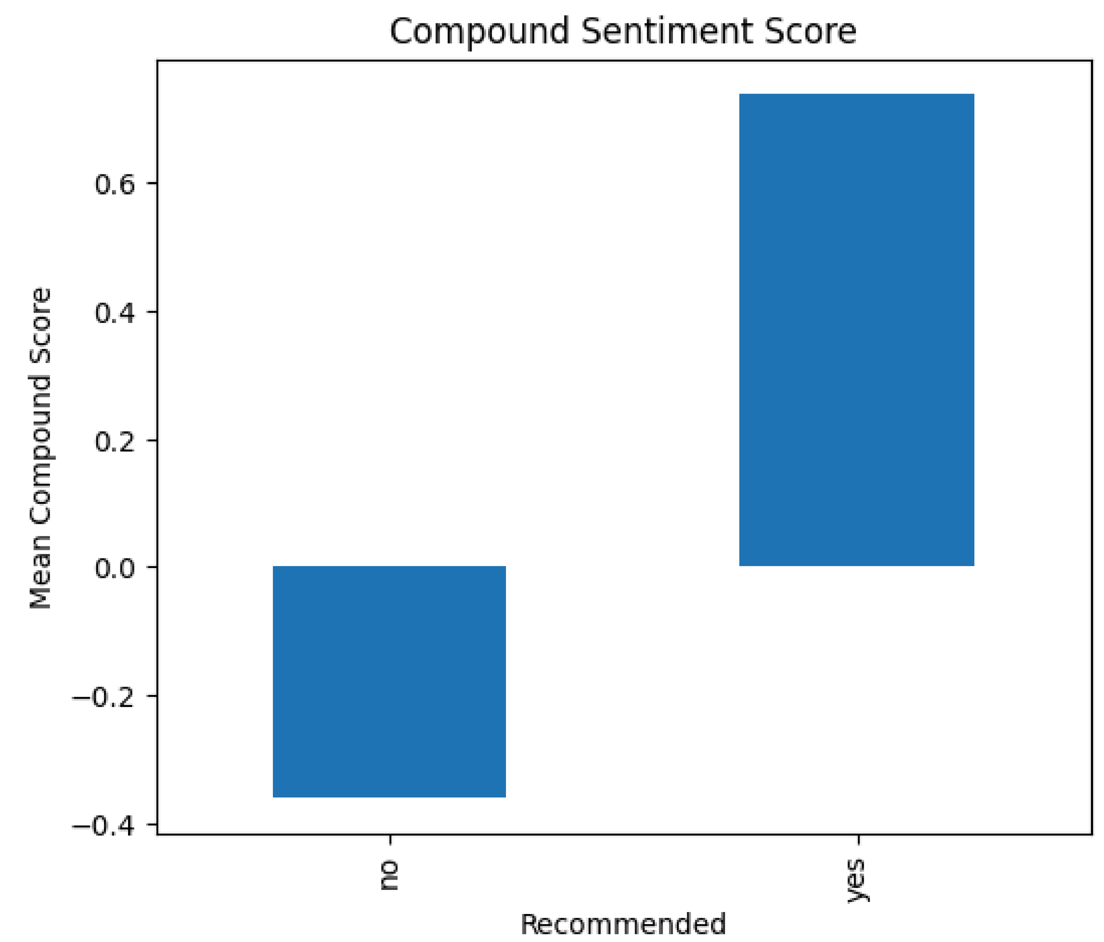

Figure 1: Sentiment by Recommendation

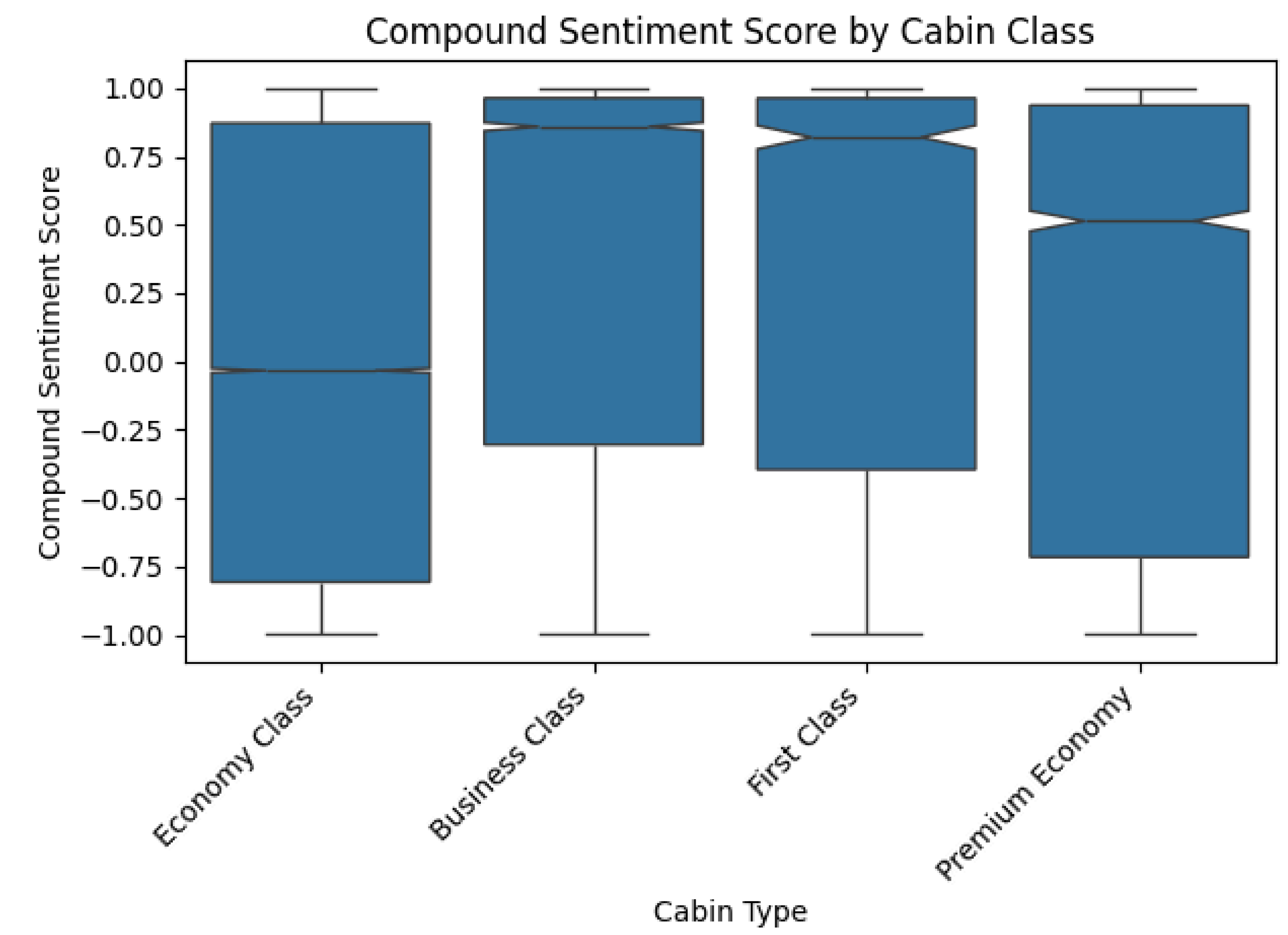

Figure 3: Sentiment by Cabin Class

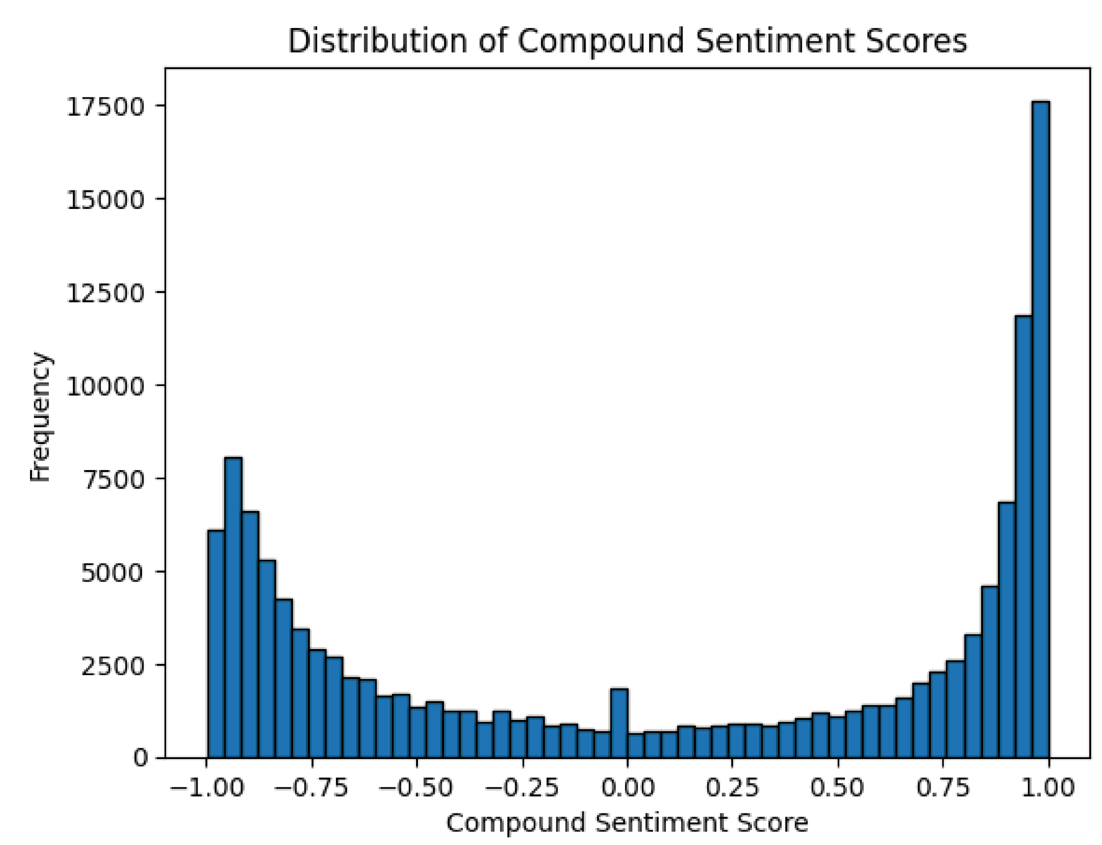

Figure 2: Sentiment Distribution

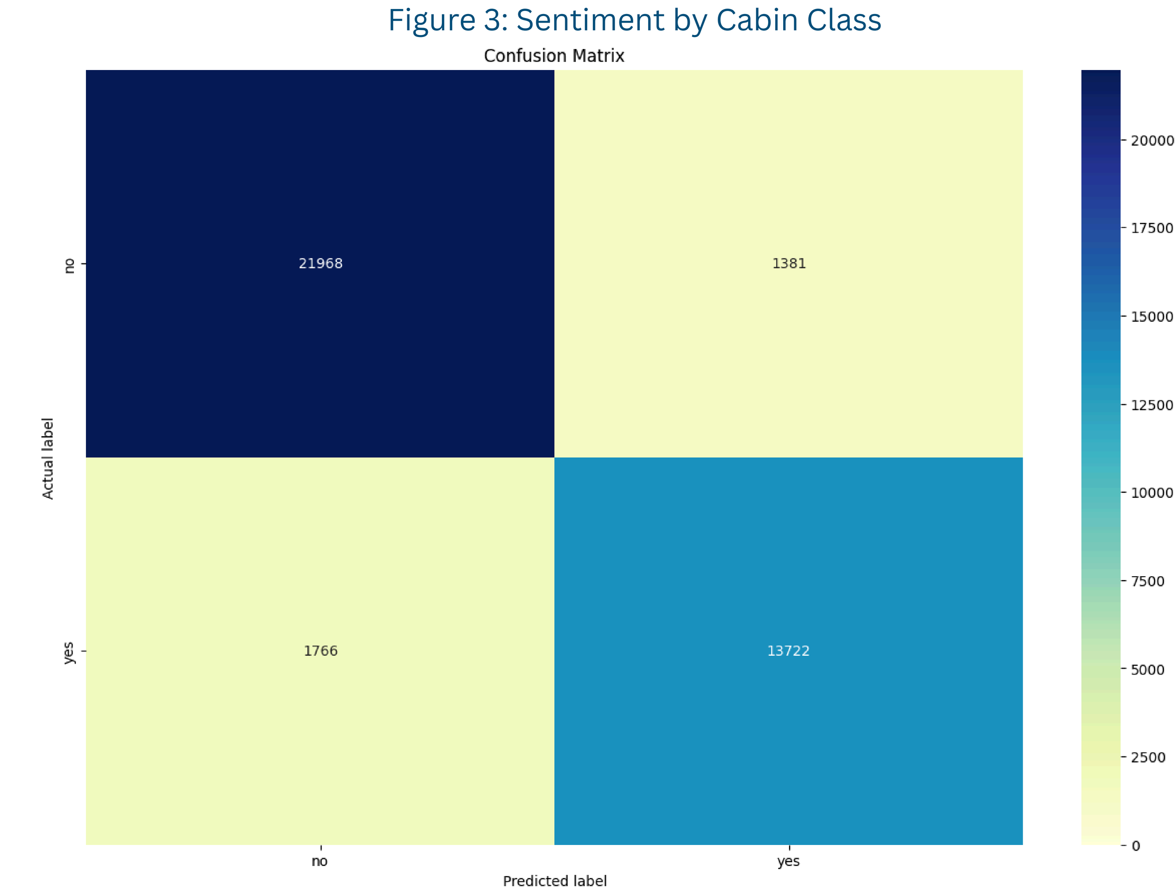

Figure 4: Confusion Matrix - Logistic Regression

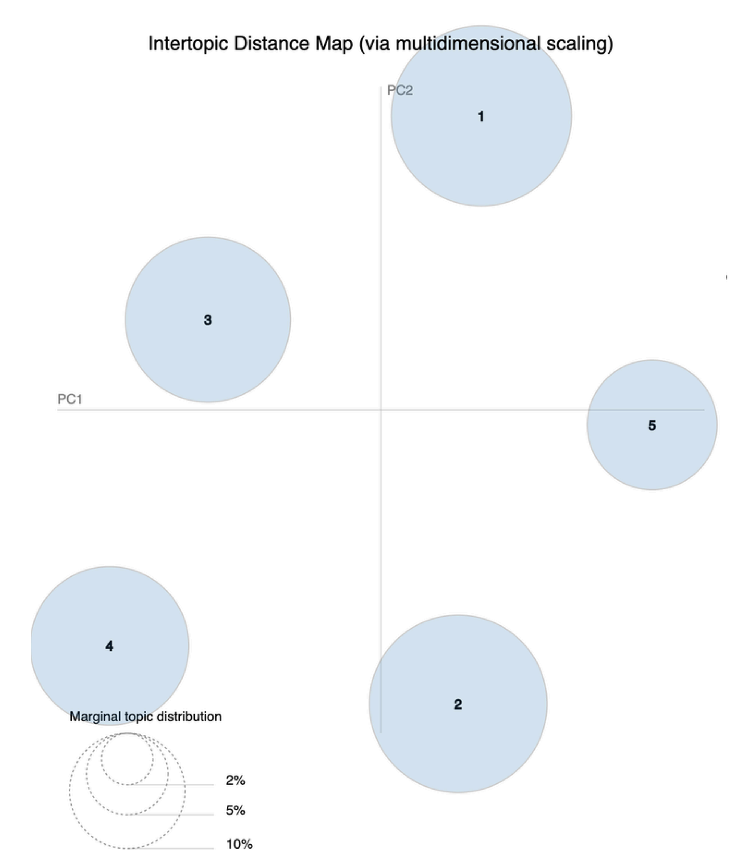

Figure 5: Intertopic Distance Map

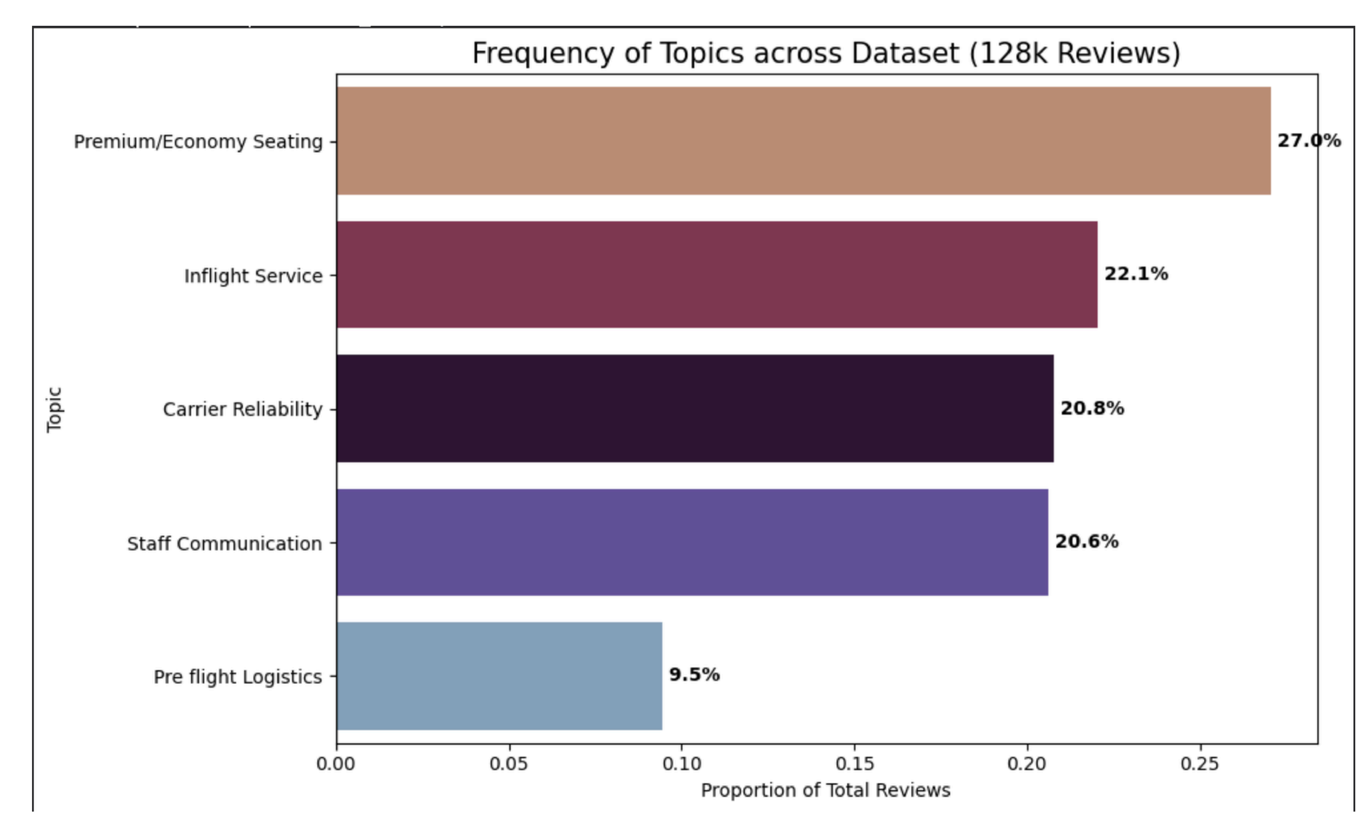

Figure 6: Topic Frequency Across Reviews

Method Selection:

Why not other methods?

- Network analysis
  - Focuses on relationships between words instead of the interpretation of the text
- Descriptive Analysis
  - Good for basic understanding
  - Does not have capture emotion

Dataset

GitHub

Findings:

Sentiment Analysis Results:

	Non Recommended	Recommended
Negative	0.101	0.039
Nautral	0.834	0.781
Positive	0.056	0.176
Compound	-0.361	0.738

Key Results:

- Recommended reviews used x3 more positive words
- Compound Sentiemnt shows clear separation in satisfied and dissatisfied customers

Cabin Type affects sentiment score:

- Business Class had the highest average sentiment score
- Economy Class had the widest range of sentiemnt scores and the lowest median

Topic Modeling Results:

Model Configuration	Coherence Score (Cv)
5 Topic Model **(Best)**	0.425
7 Topic Model	0.405
10 Topic Model	0.393

Machine learning Results:

Models:	Accuracy:
Naive Bayes	85.93%
Logistic Regression	91.90%
Random Forest	87.52%

Best Model: Logistic Regression

- Highest Accuracy: 91.90%
- Strongest Prediction and recall for recommendation prediction
- Text alone provides strong prediction

Conclusion:

The project demonstrates that airline review texts are strongly correlated with customer satisfaction. The Sentiment analysis method showed a distinct difference between the recommended and non-recommended airlines, with noticeable gaps in compound sentiment scores. Out of the three models; 5-topic model had best coherence (0.425) and managed to create topic groups that were distinct, while not being niche. The results show preprocessing was successful. Machine learning classification got an accuracy of over 91% using the Logistic Regression model. This proved that text review alone can be reliable at predicting airline recommendations. Overall, these results demonstrate how dependable sentiment monitoring and customer review detection are, and allow improvement across the airline industry.

References:

Junstorm, J (n.d.). 128k Airline Reviews. Kaggle. https://www.kaggle.com/datasets/joelljungstrom/128k-airline-reviews